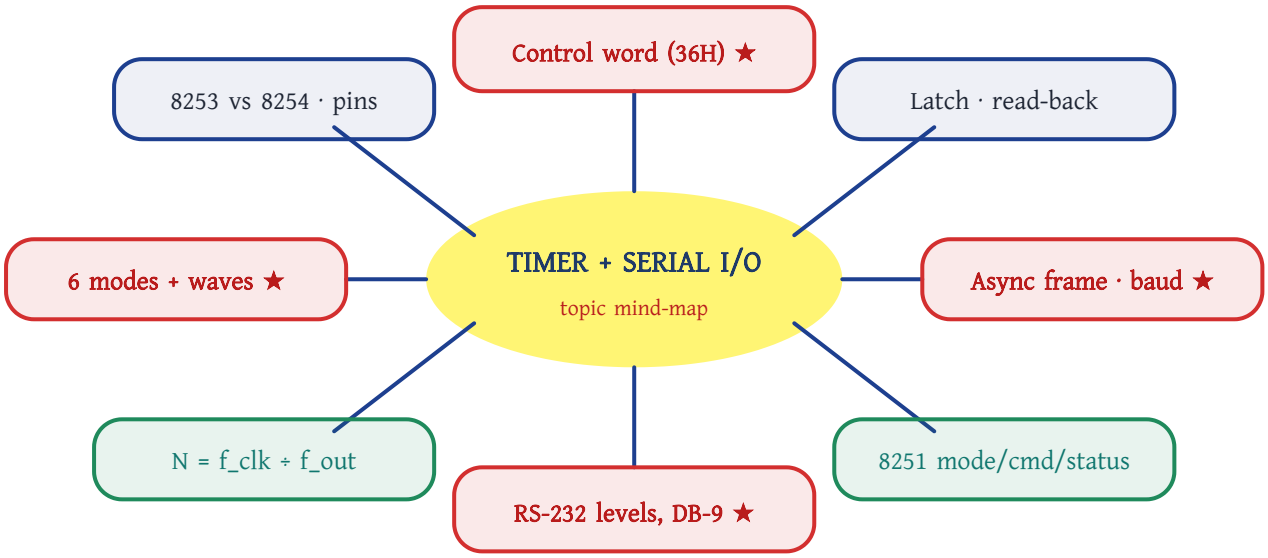
8254 Timer, Serial I/O, RS-232, 8251

PIT modes · control word · counts · async frame · RS-232 · 8251: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = calculation / programming

Syllabus coverage

"Timer interfacing: **The 8254 programmable interval timer (PIT)**, timing applications; Serial I/O interface: **asynchronous and synchronous communication**, physical communication standard-**EIA RS232**, programmable communication interface, interfacing serial I/O devices-mouse, modem, PC Keyboard". The programmable communication interface here is the **8251A USART**. The 8255A PPI and port decoding are in lesson mp2; interrupts (8259A) and DMA (8237) in mp4.

High-yield list

① 3 × 16-bit, 6 modes
② CW: SC RW M BCD
③ Mode 3 = square
④ N = f_clk ÷ f_out
⑤ RS-232: 1 = -ve
⑥ 8251: 4EH, 37H

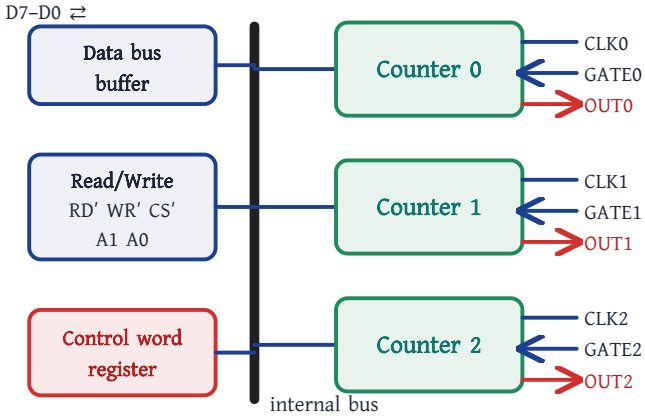
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ 8254: features, pins, vs 8253 ..... p. 2
✓ Control word & reading counters ... p. 3
✓ The six modes (waveforms) ..... p. 4
✓ Count maths & programming ..... p. 5
✓ Serial basics: frame, baud, duplex ... p. 6
✓ EIA RS-232 standard ..... p. 7
✓ 8251 USART: architecture ..... p. 8
✓ 8251 mode, command, status ..... p. 9
✓ Night-before sheet ..... p. 10
✓ Practice MCQs + written ..... p. 11

1 The 8254 Programmable Interval Timer

Definition: The Intel 8254 PIT is a 24-pin, +5 V counter/timer chip with **three independent 16-bit down counters** (Counter 0, 1, 2). Each counter has a **CLK** input, a **GATE** input and an **OUT** output, works in any of **6 modes** (0–5) and counts in **binary or BCD**, decrementing on each **falling CLK edge**. It replaces software delay loops, so the CPU is free while time is measured in hardware.



8254 block diagram

24 pins at a glance

- **D7–D0**: 8-bit data bus (8 pins)
- **CLK, GATE, OUT** × 3 counters (9 pins)
- **RD′, WR′, CS′** (active low), **A1, A0**
- V_{CC}, **GND** → total **24**

A1	A0	Selects
0	0	Counter 0
0	1	Counter 1
1	0	Counter 2
1	1	Control word reg.

8253 vs 8254 (the favourite comparison)

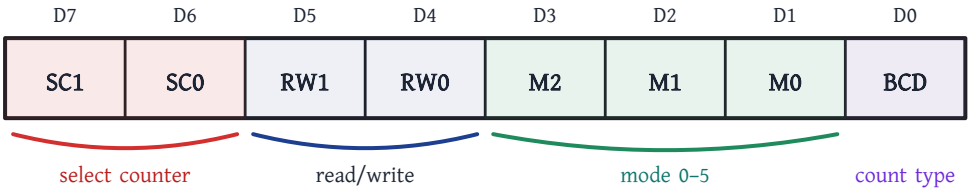
Feature	8253	8254
Maximum CLK input	**2.6 MHz**	8 MHz; **10 MHz** (8254-2)
Read-back command	no	yes (latch count + status)
Status byte of a counter	no	yes
Counters · modes · pins	3 · 6 · 24	3 · 6 · 24 (pin-compatible)

Inside each counter: **CR** (count register, written by the CPU) → **CE** (16-bit counting element) → **OL** (output latch, read by the CPU). The count moves CR → CE on the **next CLK pulse** after it is written.

Memory trick: "**3-16-6-24**": 3 counters, 16 bits each, 6 modes, 24 pins. 8254 = 8253 + **read-back** + faster clock.

Exam trap: The control word register is **write-only**; A1A0 = 11 never reads a count. Max count is loaded as **0000** = 65,536 (binary) or 10,000 (BCD).

2 The Control Word (A1A0 = 11)



SC1 SC0	Counter	RW1 RW0	Access	M2 M1 M0	Mode
0 0	Counter 0	0 0	latch command	0 0 0	0 interrupt on TC
0 1	Counter 1	0 1	LSB only	0 0 1	1 one-shot
1 0	Counter 2	1 0	MSB only	X 1 0	2 rate generator
1 1	read-back (8254)	1 1	LSB, then MSB	X 1 1	3 square wave
D0 = BCD bit: **0** = 16-bit binary (max 65,536)				1 0 0	4 soft strobe
1 = 4-decade BCD (max 10,000)				1 0 1	5 hard strobe

Worked examples: build the word ✍️:

1. Counter 0, LSB then MSB, mode 3, binary → 00 11 011 0 = **36H**
2. Counter 2, LSB then MSB, mode 0, binary → 10 11 000 0 = **B0H**
3. Counter 1, LSB then MSB, mode 3, BCD → 01 11 011 1 = **77H**
4. Decode **74H** = 01 11 010 0 → counter 1, LSB/MSB, **mode 2**, binary
5. PC speaker: counter 2, LSB/MSB, mode 3, binary → 10 11 011 0 = **B6H**

3 Reading a Counter While It Runs

1. **Simple read**: stop the count first (GATE = 0 or block CLK), then read LSB, MSB
2. **Counter latch command**: SC bits + **RW = 00** freezes CE into OL; counting continues (latch counter 1 = **40H**)
3. **Read-back** (8254 only): latches count and/or **status** of several counters at once

Read-back word:
1 1 COUNT′ STATUS′ CNT2 CNT1 CNT0 0
(COUNT′/STATUS′ = 0 → latch it)
Status byte: OUT · NULL COUNT · RW1 RW0 · M2 M1 M0 · BCD

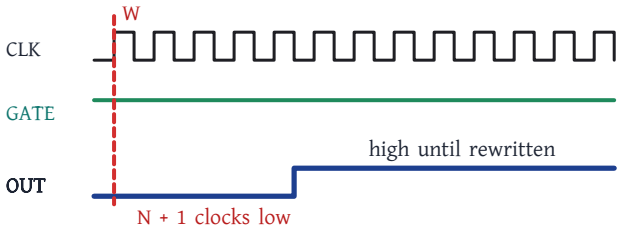
Read-back examples ✍️: latch count + status of counter 1 → 11 0 0 0 1 0 0 = **C4H** · latch status only of all three → 11 1 0 1 1 1 0 = **EEH** · latch counts of counters 0 and 2 → 11 0 1 1 0 1 0 = **DAH**

Memory trick: "**Who · How · What · Which number**": counter, access, mode, binary/BCD = 2 + 2 + 3 + 1 bits.

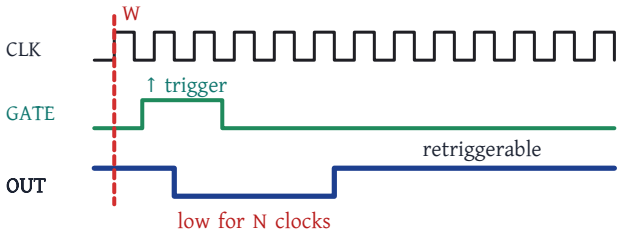
Exam trap: Mode 2 = X10, mode 3 = X11 (so 110 is mode 2). RW = 00 is **not** "no access": it is the latch command.

4 The Six Modes with Output Waveforms

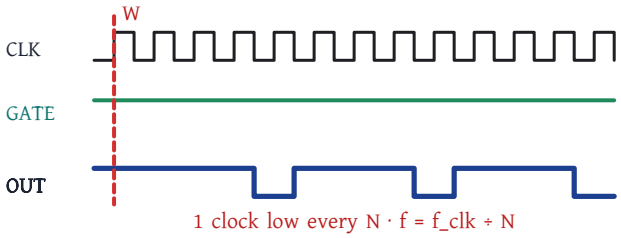
All sketches use count **N = 4**; the red dashed line marks the moment the count is written (W). Count loads on the next falling CLK edge.



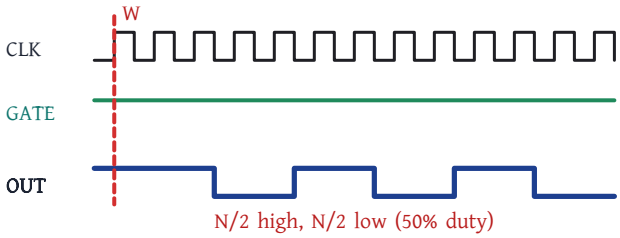
Mode 0: interrupt on terminal count



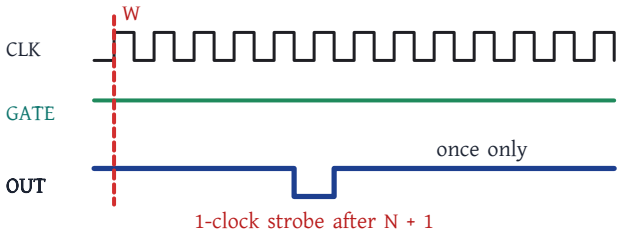
Mode 1: hardware-retriggerable one-shot



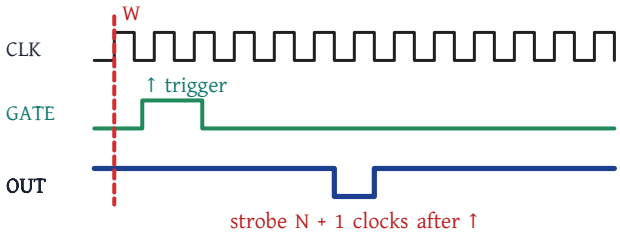
Mode 2: rate generator (÷N, auto-reload)



Mode 3: square-wave generator ★



Mode 4: software-triggered strobe



Mode 5: hardware-triggered strobe

Mode	GATE low	GATE rising edge	Repeats?	Typical use
0	pauses count	–	no	delay → interrupt
1	–	**starts / restarts**	per trigger	monostable pulse
2	stops, OUT high	reloads count	**yes, auto-reload**	tick, baud ÷N
3	stops, OUT high	reloads count	**yes, auto-reload**	tone, baud clock
4	pauses count	–	no	software strobe
5	–	**starts / restarts**	per trigger	delayed strobe

Memory trick: pairs: **0 & 4** = software, once · **1 & 5** = GATE-triggered · **2 & 3** = periodic. Adding 4 turns a long output into a **1-clock strobe**.

Exam trap: Mode 2 (narrow pulse) and Mode 3 (square wave) give the **same frequency** f_clk ÷ N; only the duty cycle differs. Minimum count in modes 2 and 3 is **2**.

5 Count Calculation & Programming

Formulas:

Modes 2, 3: **N = f_clk ÷ f_out**

Delay (modes 0, 1, 4, 5): N = t × f_clk

T_clk = 1 ÷ f_clk (2 MHz → 0.5 µs)

Lowest f_out = f_clk ÷ 65,536

Mode 3, odd N: high (N+1)/2, low (N−1)/2

N > 65,536 → **cascade two counters**

Example 1: the classic program ✍️: Counter 0 at port 40H, control at 43H, CLK0 = 2 MHz, want 5 kHz square wave.

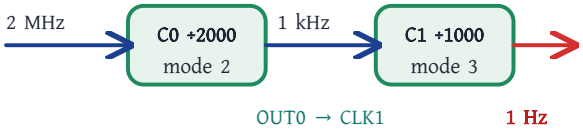
1. N = 2 MHz ÷ 5 kHz = 400 = **0190H** (for 1 kHz: **2000 = 07D0H**)
2. CW: 00 11 011 0 = **36H** (counter 0, LSB/MSB, mode 3, binary)
3. Write CW to 43H, then LSB 90H, then MSB 01H to **40H**

```
MOV AL, 36H   ; control word
OUT 43H, AL
MOV AL, 90H   ; LSB of 0190H
OUT 40H, AL
MOV AL, 01H   ; MSB of 0190H
OUT 40H, AL
```

Example 2: delay ✍️: 1 ms delay on counter 2, mode 0, CLK = 2 MHz: N = 1 ms × 2 MHz = **2000** (07D0H); CW = 10 11 000 0 = **B0H**. OUT2 goes high after 2000 + 1 clocks ≈ 1 ms → wire it to an interrupt input (8259A, lesson mp4).

Example 3: BCD ✍️: CLK1 = 1 MHz, 1 kHz in BCD: N = 1000 → write **00H** then **10H** (BCD digits 1 0 0 0); CW = 01 11 011 1 = **77H**.

Example 4: cascade ✍️: 1 Hz from 2 MHz needs N = 2,000,000 > 65,536, so one counter cannot do it. Counter 0, mode 2, N = **2000** → 1 kHz; feed OUT0 into CLK1; counter 1, mode 3, N = **1000** → **1 Hz**. Check: 2000 × 1000 = 2 × 10^6 ✓



two counters in series multiply the divisors

IT link: the IBM PC: The PC's 8253/8254 sits at ports **40H–43H** with CLK = **1.193182 MHz**. Counter 0, mode 3, count 0 (65,536) → **18.2 Hz** system tick on IRQ0 (every 54.9 ms); counter 1, mode 2, count 18 (CW 54H) → DRAM refresh every 15.1 µs; counter 2, mode 3 (CW B6H) → speaker tone (1 kHz needs N ≈ 1193).

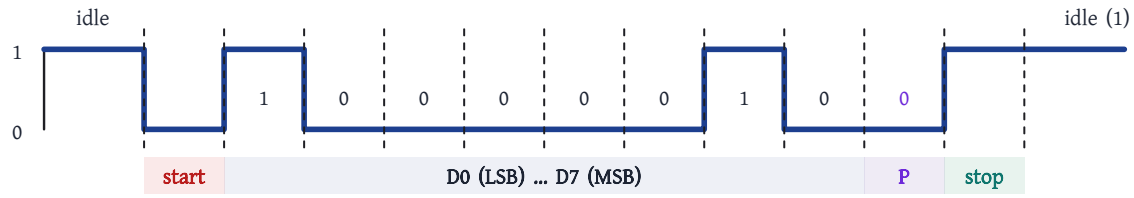
Exam trap: N is a **frequency ratio**; do not divide periods the wrong way round. Write the count to the **counter's** port, LSB first, never to the control port.

6 Serial vs Parallel; Synchronous vs Asynchronous

	Serial	Parallel
Bits/clock	1	8 or more
Distance	**long**	short (skew)
Wires	few	many
Cost	low	high
Example	RS-232	8255A

	Asynchronous	Synchronous
Clock	none (agreed baud)	shared / SYNC
Framing	**start + stop / char**	SYNC per **block**
Overhead	high (8N1: 20%)	low
Speed	lower	higher
Example	UART, mouse	HDLC, SPI

7 The Asynchronous Character Frame



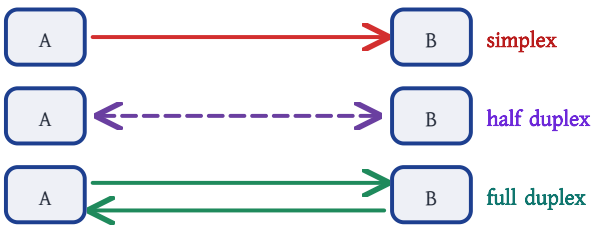
'A' = 41H = 0100 0001 · even parity → P = 0 · 1 + 8 + 1 + 1 = 11 bits

- Idle line = **1 (mark)**; **start bit = 0** (1 bit, the 1→0 edge wakes the receiver)
- Data: **5–8 bits, LSB first**
- Parity: even / odd / none (optional)
- Stop bit(s) = 1: **1, 1.5 or 2** bits

Baud vs bit rate: **Baud** = signal changes (symbols) per second; **bit rate** = baud × bits per symbol. With 2-level signalling (RS-232, UART) **baud = bps**.

Worked examples ✍️: ① 9600 baud, 8N1 (1 + 8 + 1 = 10 bits) → **960 chars/s**; bit time = 1 ÷ 9600 = 104.17 µs · ② 9600 baud, 8 data + parity + 2 stop = 12 bits → **800 chars/s** · ③ modem at 2400 baud with 16 levels (4 bits/symbol) → 2400 × 4 = **9600 bps** · ④ efficiency of 8N1 = 8 ÷ 10 = **80%**

8 Simplex, Half Duplex, Full Duplex

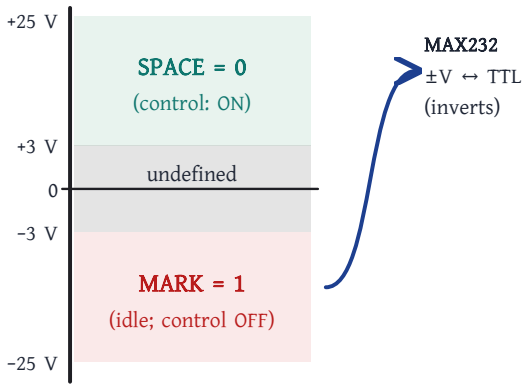


- **Simplex**: one way only (broadcast radio/TV, keyboard → CPU)
- **Half duplex**: both ways, **one at a time** (walkie-talkie)
- **Full duplex**: both ways **at the same time** (telephone; RS-232 with TxD + RxD)

Memory trick: The frame "**starts low, stops high**": start = 0, stop = 1, so every character begins with a guaranteed 1 → 0 edge.

9 EIA RS-232: the Physical Standard

Definition: RS-232 (EIA "Recommended Standard", 1962; RS-232-C 1969) defines the **electrical, mechanical and functional** interface between a **DTE** (data terminal equipment: PC) and a **DCE** (data circuit-terminating equipment: modem). Single-ended, point-to-point, **negative logic** on data lines.

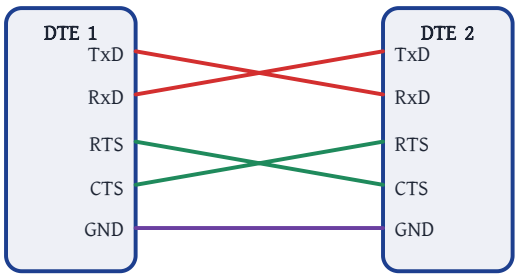


RS-232 data-line levels

Parameter	Value
Logic 1 (mark)	**−3 to −25 V**
Logic 0 (space)	**+3 to +25 V**
Driver (typ.)	±5 to ±15 V (±12 V)
Max length	≈ 15 m (50 ft)
Max rate	20 kbps (spec)
Connector	DB-25, DB-9 (PC)

TTL ↔ RS-232: **MAX232** (single +5 V, charge pumps) or MC1488 driver + MC1489 receiver.

Signal (DTE view)	DB-9	DB-25
TxD transmit data	3	2
RxD receive data	2	3
RTS request to send	7	4
CTS clear to send	8	5
DSR data set ready	6	6
Signal ground	**5**	**7**
DCD carrier detect	1	8
DTR data terminal ready	4	20
RI ring indicator	9	22



null modem: cross TxD↔RxD, RTS↔CTS, DTR↔DSR

Handshake: DTR/DSR = "ready"; RTS/CTS = "may I send?" / "go ahead" (flow control); DCD = carrier; RI = ringing. A **null modem** links two DTEs (minimum: TxD, RxD crossed + GND).

Standard	Signalling	Topology	Max distance	Max rate
RS-232	single-ended	1 driver, 1 receiver	≈ 15 m	20 kbps
RS-422	**differential**	1 driver, 10 receivers	1200 m	10 Mbps
RS-485	differential	**multi-drop**, 32 nodes	1200 m	10 Mbps

Exam trap: TxD is pin **3** on DB-9 but pin **2** on DB-25; ground is 5 vs 7. Data lines use negative logic, but control lines are **asserted (ON) when positive**.

Memory trick: "**Mark is minus**": both start with M. A line idling at about −12 V is sending 1s.

10 8251A USART (Communication Interface)

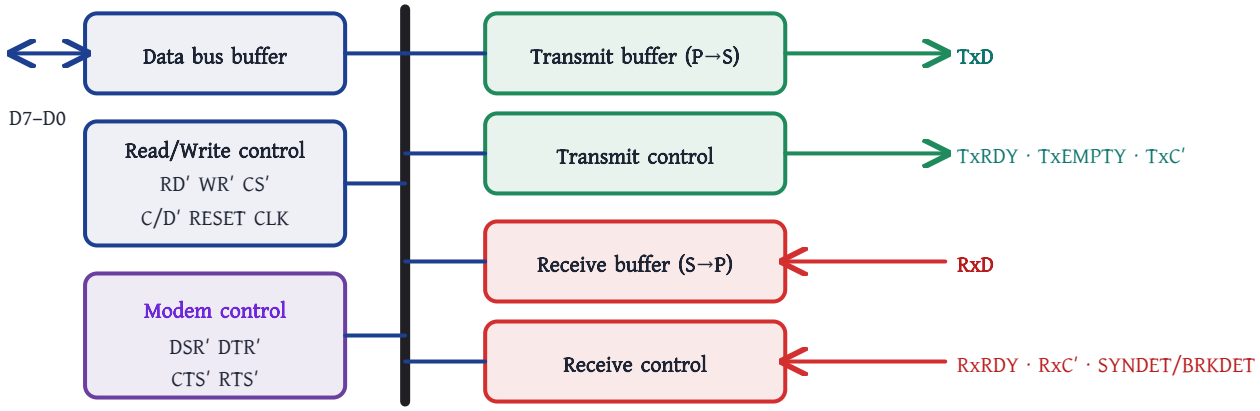
Definition: The Intel 8251A **Universal Synchronous/Asynchronous Receiver/Transmitter** (28 pins, +5 V) turns CPU bytes into serial bits (**P → S**, TxD) and serial input into bytes (**S → P**, RxD); full duplex.

- Async: **5–8 data bits**, clock ×1, ×16 or ×64, 1/1.5/2 stop bits, up to **19.2 kbps**
- Sync: one or two SYNC characters, internal or external sync detect, up to **64 kbps**
- Parity even/odd/none; detects parity, overrun, framing errors and break

UART vs USART:

UART = async only (8250, **16550**: 16-byte FIFO, PC COM ports)

USART = both modes (**8251A**)



C/D′	RD′	WR′	Operation
0	0	1	CPU reads **data**
0	1	0	CPU writes **data**
1	0	1	CPU reads **status**
1	1	0	CPU writes **mode / command**

- **TxRDY**: CPU may write the next byte
- **TxEMPTY**: last bit has left TxD
- **RxRDY**: a byte is waiting for the CPU
- TxRDY/RxRDY can drive IRQs (8259A)

Interfacing serial I/O devices

Device	How it is interfaced
Serial mouse	async RS-232, classically **1200 baud, 7N1**, 3-byte packets
Modem (DCE)	async RS-232 + full handshake; digital ↔ analog; AT commands
PC keyboard	**synchronous serial** (CLOCK + DATA), 11-bit frame, scan codes → 8042, port 60H, IRQ1

Memory trick: C/D′ = "**C**ontrol when 1, **D**ata when 0". Control + read = status; control + write = mode/command.

Exam trap: The 8251 has **no internal baud generator**: TxC/RxC must come from outside, typically an 8254 in mode 3.

11 Mode, Command and Status Words

Mode word (asynchronous): written first after reset

D7 D6: S2 S1	D5: EP	D4: PEN	D3 D2: L2 L1	D1 D0: B2 B1
stop bits	parity type	parity enable	data bits	baud factor
00 invalid · 01 = 1	1 = even	1 = on	00 = 5 · 01 = 6	00 = **sync** · 01 = ×1
10 = 1.5 · 11 = 2	0 = odd	0 = none	10 = 7 · 11 = 8	10 = ×16 · 11 = ×64

Sync mode word: B2B1 = 00; D7 = **SCS** (1 = one SYNC char), D6 = **ESD** (1 = external sync).

Worked examples ✍️: ① 8 data, no parity, 1 stop, ×16 → 01 0 0 11 10 = **4EH** · ② 7 data, even parity, 2 stop, ×64 → 11 1 1 10 11 = **FBH** · ③ 7 data, odd parity, 1 stop, ×16 → 01 0 1 10 10 = **5AH** · ④ decode **7EH** = 01 1 1 11 10 → 1 stop, even parity, 8 data, ×16

Command word: every later control write

D7	D6	D5	D4	D3	D2	D1	D0
EH	**IR**	**RTS**	**ER**	**SBRK**	**RxE**	**DTR**	**TxEN**
hunt	reset	RTS′ = 0	clear errors	break	Rx on	DTR′ = 0	Tx on

Worked examples ✍️: RTS + ER + RxE + DTR + TxEN → 0011 0111 = **37H** · ER + RxE + TxEN → **15H** · internal reset → **40H**

Status word: read with C/D′ = 1

D7	D6	D5	D4	D3	D2	D1	D0
DSR	SYNDET	**FE** framing	**OE** overrun	**PE** parity	TxEMPTY	RxRDY	TxRDY

Decode ✍️: 85H = 1000 0101 → DSR, TxEMPTY, **TxRDY**: free to send · 1AH = 0001 1010 → **OE + PE + RxRDY**: byte with parity error, an earlier byte lost

Initialise and poll (data 80H, control 81H)

1. Reset: write **00H three times, then 40H** (internal reset) to the control port
2. **Mode word** (e.g. 4EH); in sync mode follow it with 1 or 2 SYNC characters

3 **Command word** (e.g. 37H), then poll status and move data

```
TX: IN   AL, 81H ; status
    TEST AL, 01H ; TxRDY?
    JZ   TX
    MOV  AL, BL
    OUT  80H, AL ; send
```

```
RX: IN   AL, 81H ; status
    TEST AL, 02H ; RxRDY?
    JZ   RX
    IN   AL, 80H ; byte in
```

Memory trick: From D0 up: "**T-R-T**" (TxRDY, RxRDY, TxEMPTY), then errors "**P-O-F**".

Exam trap: Baud = TxC ÷ factor: 9600 baud at ×16 needs TxC = **153.6 kHz**. FE exists in **async** mode only.

Night-Before-the-Exam Sheet

Fact sheet:

8254: 3 × 16-bit, 6 modes, 24 pins

8253 2.6 MHz · 8254-2 10 MHz

A1A0 = 11 → control word (write-only)

CW = SC1 SC0 RW1 RW0 M2 M1 M0 BCD

RW 00 = latch · 11 = LSB then MSB

Read-back (8254 only): 11 C′ S′ C2 C1 C0 0

N = f_clk ÷ f_out · count 0 = 65,536

36H = C0, LSB/MSB, mode 3, binary

Frame: 0 · 5–8 data, LSB first · P · 1

8N1 at 9600 baud → 960 chars/s

RS-232: 1 = −3…−25 V · 0 = +3…+25 V

DB-9: TxD 3, RxD 2, GND 5

8251 C/D′: 1 = control/status, 0 = data

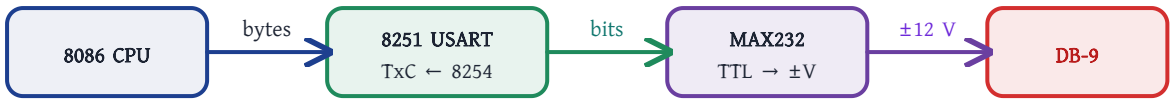
4EH = 8N1 ×16 · 37H = Tx Rx ER RTS DTR

Top 12 one-liners for MCQs

1. Mode 3 = square-wave generator
2. Mode 2 = rate generator (÷N)
3. Mode 0 = interrupt on terminal count
4. Modes 1, 5 = triggered by GATE ↑
5. Read-back command: 8254 only
6. PC timer clock 1.193182 MHz, 18.2 Hz tick
7. bps = baud × bits per symbol
8. Async = start/stop per character
9. RS-232 = negative logic, ≈ 15 m
10. MAX232: TTL ↔ RS-232 from +5 V
11. Null modem crosses TxD ↔ RxD
12. 8251 status D0 = TxRDY, D1 = RxRDY

Top traps: A1A0 = 11 is the control register, not counter 3 · RW = 00 is the latch command · LSB before MSB, to the counter port · mode 2 vs 3: same frequency, different duty · TxD = pin 3 on DB-9, pin 2 on DB-25 · mark (1) is negative · baud = bps only for 2-level signals · 8251 needs an external baud clock

All tricks together: 3-16-6-24 · "Who · How · What · Which number" for the CW · mode pairs 0&4, 1&5, 2&3 · start low, stop high · mark is minus · C/D′: Control = 1 · status bits T-R-T, then P-O-F



the complete serial port chain: parallel → serial → line levels → connector

✓ Practice MCQs (exam pattern, 10 marks)

1. Which feature does the 8254 have that the 8253 lacks?

(a) Six modes (b) BCD count
(c) 3 counters (d) Read-back

2. The 8254 control word 36H selects:

(a) C0, mode 3 (b) C1, mode 2
(c) C2, mode 3 (d) C2, mode 0

3. In the 8254, A1A0 = 11 selects:

(a) Counter 0 (b) Counter 2
(c) Control reg. (d) Status reg.

4. 8254 mode for a square wave:

(a) Mode 2 (b) Mode 3
(c) Mode 1 (d) Mode 5

5. 2 MHz CLK, 4 kHz square wave: N = ?

(a) 2000 (b) 400
(c) 250 (d) 500

6. In RS-232, logic 1 (mark) is:

(a) +3 to +25 V (b) −3 to −25 V
(c) 0 to +5 V (d) −3 to +3 V

7. On a PC's DB-9 connector, TxD is pin:

(a) 2 (b) 3
(c) 5 (d) 7

8. 9600 baud, 1 start + 8 data + parity + 2 stop: chars/s?

(a) 960 (b) 1200
(c) 800 (d) 873

9. 8251 mode word 4EH means:

(a) 8N1, ×16 (b) 7E2, ×64
(c) 8O1, ×1 (d) sync, 8 bits

10. 8251 status bit: received byte waiting?

(a) TxRDY (b) TxEMPTY
(c) SYNDET (d) RxRDY

Answer key → 1-d · 2-a · 3-c · 4-b · 5-d · 6-b · 7-b · 8-c · 9-a · 10-d

Written question: 8254 counter 0 (port 80H, control 83H, CLK0 = 1.8432 MHz) must clock an 8251 (data 90H, control 91H) for 9600 baud at ×16. Program both (8251: 8 data, even parity, 1 stop, Tx + Rx on). 8 marks

1. TxC = 9600 × 16 = 153.6 kHz → N = 1,843,200 ÷ 153,600 = **12 = 000CH** 2
2. 8254 CW: counter 0, LSB/MSB, mode 3, binary = **36H** 2
3. 8251 mode: 01 1 1 11 10 = **7EH** (1 stop, even parity, 8 bits, ×16) 2
4. 8251 command: **37H** (TxEN, DTR, RxE, ER, RTS); program below 2

```
MOV AL, 36H
OUT 83H, AL   ; 8254 CW
MOV AL, 0CH
OUT 80H, AL   ; LSB
MOV AL, 00H
OUT 80H, AL   ; MSB
```

```
MOV AL, 7EH
OUT 91H, AL   ; 8251 mode
MOV AL, 37H
OUT 91H, AL   ; command
```

Before the exam ✓

- ✓ Build/decode any 8254 control word
- ✓ Sketch OUT for all six modes
- ✓ Find N; cascade when N > 65,536

Also practise (written):

① 8254 at 2 MHz: 1 ms delay

② Latch and read counter 1

③ 8251 receive with error check